VishGuard: Defending against Vishing

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***Abstract*—Vishing and smishing attacks, or voice phishing attacks, are a type of social engineering attack in which attackers use voice communication channels, such as phone calls or text messages, to trick victims into divulging sensitive information or performing actions that compromise their security. Vishing attacks have become increasingly common and sophisticated in recent years and have had severe consequences for individuals, businesses, and organizations. In this review, we introduce VishGuard, a comprehensive framework designed to detect and mitigate vishing attacks using advanced natural language processing (NLP) and machine learning techniques. VishGuard leverages Python and Streamlit to create a multi-functional application capable of analyzing and processing voice data to identify potential vishing threats.**

***Keywords*: vishing, voice phishing, social engineering, algorithm, tokenization, natural language processing, machine learning**

I. Introduction

In an era where digital communication is ubiquitous, the threat of scams, particularly vishing (voice phishing) and other forms of social engineering attacks, has become increasingly sophisticated and pervasive. These malicious activities exploit various channels including text messages, emails, voice calls, and even QR codes, posing significant risks to individuals, businesses, and organizations [1]. One way for wireless communication security is to implement integration approaches to execute security schemes in wireless protocols with high performance presented in [2]. On the other hand, the contents of digital communication need to be examined.. However, the existing security measures have struggled to keep pace with the evolving tactics of scammers, leaving many vulnerable to monetary loss and data breaches. In response to these evolving threats, we present VishGuard, a web-based scam detection platform that leverages algorithms and machine learning to identify and mitigate several types of scams across different communication channels.

The rationale behind VishGuard's multi-faceted approach lies in the understanding that modern scams are not limited to a single communication channel or technique [3]. By combining various detection methods and data sources, VishGuard aims to provide a robust and adaptable defense against a wide range of scam attempts. This comprehensive strategy addresses the complex and evolving nature of modern scam techniques.

In the following sections of this paper, we will delve deeper into the technical architecture of VishGuard, discuss our methodology in detail, present our findings, and explore the implications and future directions of this technology in the ongoing battle against scams and social engineering attacks. Through this comprehensive examination, we will demonstrate how VishGuard represents a significant step forward in cybersecurity defense strategies, offering a dynamic and responsive solution to the persistent threat of scams and reducing the amount of phishing scams faced [4].

II. Literature Review

Vishing has emerged as a significant cybersecurity threat in recent years. Several studies have examined this form of social engineering attack and proposed countermeasures.

A. Current Landscape of Vishing Research

The landscape of digital threats has evolved rapidly in recent years. [5] conducted a comprehensive analysis of these attacks, revealing their increasing sophistication and the significant challenges they pose to detection efforts. Delving deeper into the human aspect of this problem, [6] investigated the psychological factors that render individuals susceptible to vishing attacks. Their psychological insight is vital for understanding why traditional security measures often fall short in protecting users.

Recognizing the need for technological intervention, [7] proposed an innovative machine learning-based approach specifically designed to identify potential vishing attempts. However, despite these valuable contributions, a significant gap remains in the field of vishing defence. While several studies have addressed specific aspects of vishing detection, such as call analysis or user behaviour, there is a notable absence of comprehensive, integrated defence systems. Most existing solutions focus on isolated aspects of the problem, leaving users vulnerable to the multi-faceted nature of modern vishing attacks.

This fragmented approach to vishing defence highlights a clear opportunity for innovation. There is a pressing need for a unified platform that can address the various dimensions of vishing attacks simultaneously, combining insights from psychological research, advanced machine learning techniques, and comprehensive data analysis. Such an integrated system could potentially offer a more robust and adaptable defence against the ever-evolving tactics of vishing scammers.

Recognizing the critical nature of this threat, governments and organizations worldwide have begun to invest in innovative solutions to combat vishing and related scams. Singapore has taken proactive steps to address this growing concern, funding the development of innovative technologies aimed at protecting its citizens from sophisticated phone scams [8]. Two notable products have emerged from this initiative, each offering unique approaches to the challenge of vishing detection and prevention.

B. ScamShield

ScamShield, an innovative anti-scam product suite, has been downloaded by more than 820,000 users and has successfully blocked over 7 million suspicious SMSs reported

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as scams. The app's primary features include automatic blocking of scam calls, automatic detection of scam messages, and user reporting of scam messages and calls. It operates by cross-referencing unknown numbers with a database maintained by the Singapore Police Force, while employing an on-device algorithm to assess messages from non-saved contacts for SMS detection [9].

To complement the app, the ScamShield Bot was recently soft launched on WhatsApp. This bot enhances the system's capabilities by allowing users to check if a message or call is a scam and report scams directly to the Singapore Police Force.

C. CheckMate SG

Checkmate SG [10] broadens the scope of anti-scam efforts by addressing not only text messages but also suspicious images and online videos [11]. Checkmate SG operates through a WhatsApp bot, allowing users to submit potentially fraudulent content for verification.

D. Comparison with VishGuard

VishGuard emerges as a new contender in the fight against digital scams, sharing commonalities with existing systems while introducing novel features. Like ScamShield and Checkmate SG, VishGuard harnesses the power of advanced Natural Language Processing (NLP) techniques and machine learning algorithms to analyze and detect potential threats. All three systems aim to provide real-time protection against scams and fraudulent activities, leveraging technological solutions to safeguard users in an increasingly complex digital landscape.

However, VishGuard distinguishes itself in several key areas. While ScamShield and Checkmate SG primarily focus on text-based scams and misinformation, VishGuard expands its scope to include voice-based threats. This is achieved through an innovative approach of transcribing audio recordings and employing a string-matching algorithm to compare the transcribed text against known vishing patterns and scripts. This capability addresses a critical gap in existing solutions, tackling the persistent threat of voice phishing that continues to evolve and challenge traditional defense mechanisms.

The development of VishGuard, alongside initiatives like ScamShield and Checkmate SG, represents a significant step forward in creating a more comprehensive defense against digital scams. By addressing voice-based threats in addition to text-based scams, VishGuard complements existing solutions and contributes to a more robust, multi-faceted approach to cybersecurity.

III. Solution Designed

A. Overview

VishGuard is designed as a comprehensive web-based application that leverages advanced technologies to detect and prevent vishing attacks. At its core, the system features a user-friendly web interface built with Streamlit, providing an intuitive platform for user interaction. This web app offers two primary functionalities: media analysis and text analysis.

The media analysis capability allows VishGuard to process and analyze various forms of media, including audio files and images, which is crucial for examining potential vishing calls and visual phishing attempts. Simultaneously, the text analysis feature enables users to input text messages for scrutiny, facilitating the detection of text-based phishing attempts. This dual-input capability ensures that VishGuard offers a versatile defense mechanism, capable of analyzing both voice-based vishing attempts, suspicious text messages and malicious links.

B. Text and Image Detection

The user can enter text or upload images that could be a screenshot of a message or a QR Code. If it is a QR image, it is first decoded using pyzbar library to obtain the text from the URL. The Knuth-Morris-Pratt (KMP) algorithm is selected for string matching due to its linear time complexity time complexity of $O(n + m)$, where n is the length of the text and m is the length of the pattern. Hence, KMP algorithm is then used to check if it begins with “http” or “https” to identify that it indeed is a link and it displays the link along with a button called “check URL safety” to the user. Upon clicking, the link stored as text is checked using Uniform Resource Locator (URL) safety from VirusTotal API to determine whether the link is malicious.

If it is not a QR image, the image is then read by PyTessaract, it is then converted to grayscale along with preprocessing to enhance text visibility. PyTesseract then analyses the image, segmenting it into blocks of text, lines, and words. It then applies OCR algorithms to identify and extract text characters.

Postprocessing: The extracted text is cleaned and formatted to remove any noise or artifacts. The extracted text is then fed into text detection and will be checked against our Machine Learning Model.

Should the user only enter text, the spam prediction function uses a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer and a pre-trained model trained with known datasets that will be explained in later sections. Before the text are analysed, we preprocess the text given by the users and remove any stop words that do not contribute to the analysis to prevent the model’s accuracy from being affected. Once done, the text will be analysed using our model while also incorporating the KMP algorithm and regular expressions to identify URLs and check it against the URL safety checks from VirusTotal API to determine whether the present links are malicious or not. If the user only includes a URL, the scam probability will be based off VirusTotal API’s URL checker to determine whether it is a scam or not.

In Fig.1., after the user has inputted a scam text, our model displays the prediction result, including the probability of scam and URL safety information. The system also includes a questionnaire for further evaluation of potential scams at all levels of probability as there could be new innovative variants of vishing available in the web, acting as a failsafe should our model fail to pick up the malicious texts. If the model reaches a 80% probability rate and above, it will be classified as a scam automatically and the user should take any precaution and dissociate themselves with the text.

C. Audio Detection

When a user submits a voice recording, the system first utilizes the Faster Whisper library to transcribe the audio into text with high accuracy and speed. The text is then tokenized to remove irrelevant stop words before undergoing analysis. Instead of using basic nested for loops to search for patterns,

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which can be inefficient and have quadratic time complexity in the worst case, the system employs the KMP algorithm. KMP offers a significant performance improvement due to its linear time complexity, O(n+m), where n is the length of the text and m is the length of the pattern.

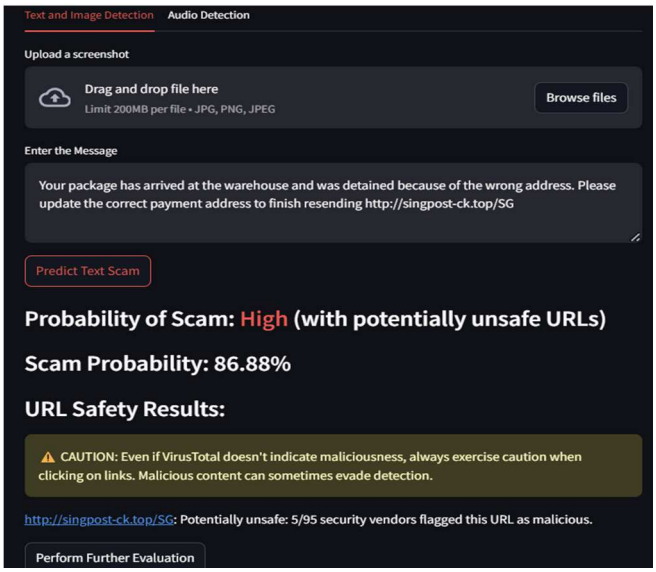

Fig. 1. Text & URL detection in action

This efficiency is achieved through the algorithm's preprocessing phase, which constructs the Longest Prefix Suffix (LPS) array to avoid redundant comparisons. While nested for loops may repeatedly compare overlapping substrings, KMP intelligently skips ahead based on previously matched characters, thus reducing the number of comparisons. This capability makes KMP particularly effective for detecting patterns in large volumes of text and for adapting to variations in vishing tactics, ensuring robust and scalable threat detection. Fig. 2 shows VishGuard transcribes and analyses audio.

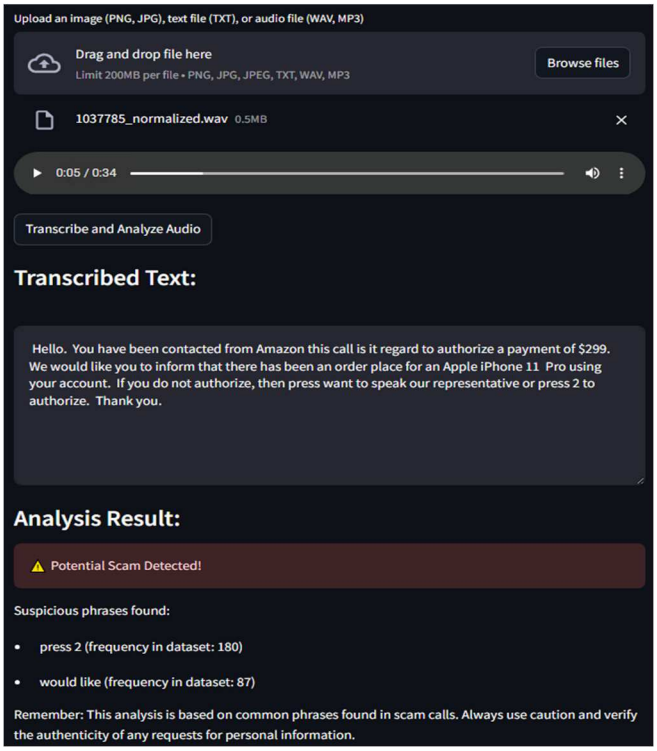

Fig. 2. Media Inspector

Regardless of whether a threat is detected, each analysed communication contributes to the continuous improvement of the system. The database update component incorporates new threat information, including both automatically detected threats and those verified by cybersecurity experts through the expert verification module. This constant influx of new data helps keep the system up to date with the latest vishing tactics and patterns.

Additionally, as can be seen in Fig. 3., if users believe message and questionnaire are still insufficient for them to determine whether the content is a scam, they can send their message for administrators to submit messages they believe were incorrectly categorized, which are stored in a postgres database, supabase an online database, for the administrators to review.

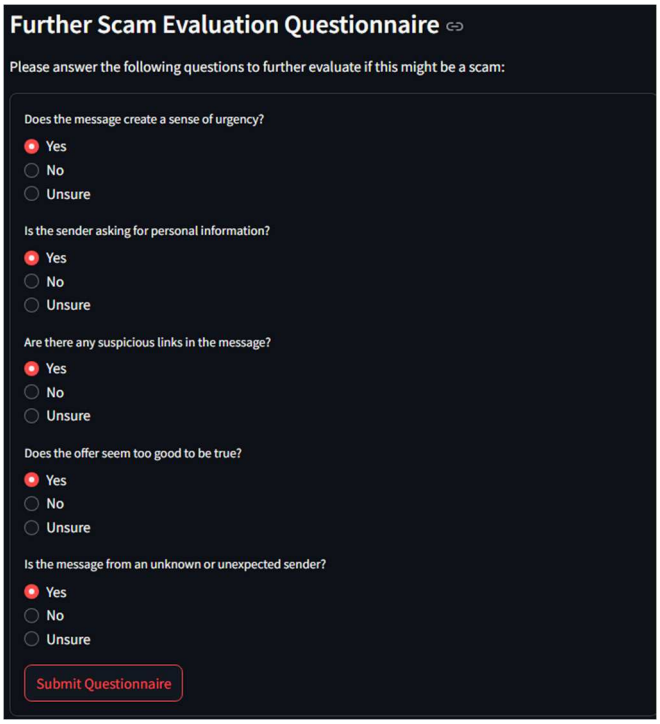

Fig. 3. Questionnaire after prediction

After which, the administrator can review the message and verify whether it is a spam or ham, a term used to define non malicious texts and as can be seen in Fig. 4.

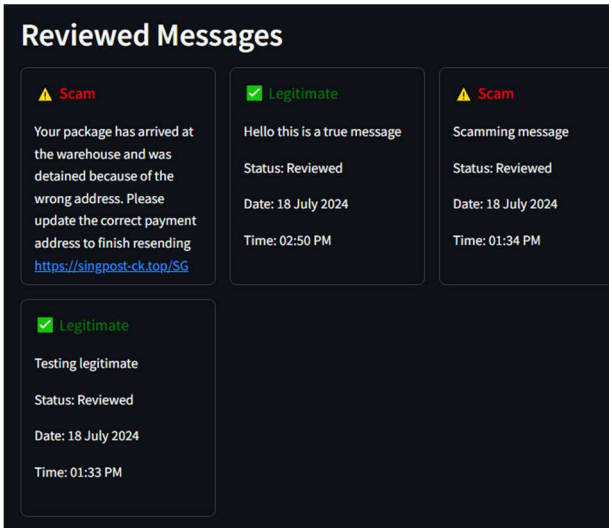

Fig. 4. Reviewed messages

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Thus, users can view whether reviewed messages are defined as malicious or not to educate themselves. We took inspiration from CheckMate regarding this feature as we believed it being beneficial to the public due to its educational significance.

As such, users of Vish Guard will have multiple failsafe to rely upon to not fall into the trap of being scammed if they were to utilize our website as well as learn more about innovative scams that may be harder to detect and thus fall prey to.

D. Admininstrator Dashboard

The data containing the collection of scam messages is retrieved from Supabase and stored in a Data Frame, which is organized in rows and columns, like a 2D matrix (Fig. 5).

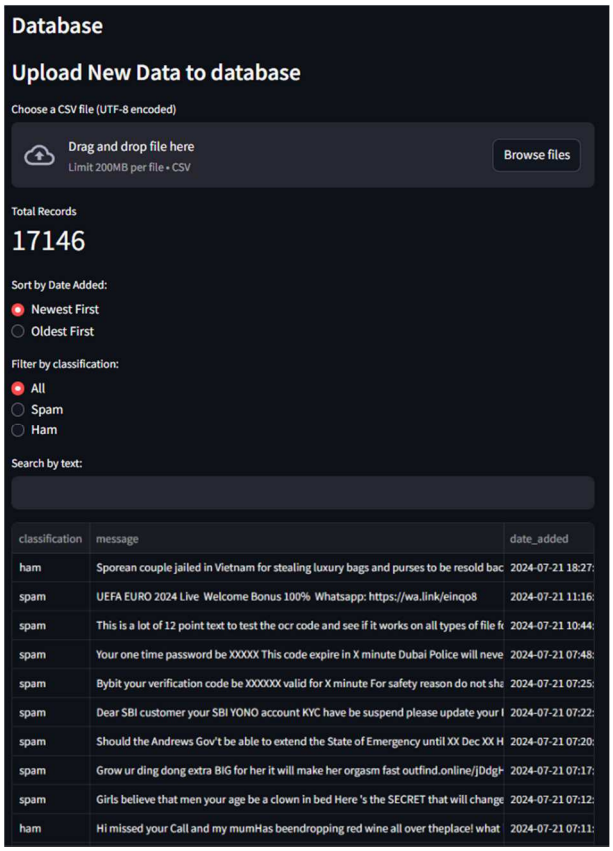

Fig. 5. Overview of database

The data upload feature allows administrators to upload CSV files that are UTF-8 encoded. The system includes an encoding validation check to ensure the uploaded file meets this requirement. If the encoding is incorrect, the system provides guidance to the administrator on how to properly encode the file. Once a valid file is uploaded and confirmed by the administrator, the system has the capability to insert this new data into the Supabase database, expanding the dataset used for scam detection.

To provide administrators with a quick overview of the data flow in the Database tab follows a clear process. First, an administrator uploads a CSV file. The system then validates the file for UTF-8 encoding. Upon confirmation from the administrator, the data is inserted into the Supabase database. The dashboard then fetches this updated data from Supabase, allowing the administrator to immediately see the new entries. From there, the administrator can sort, filter, and search the displayed data as needed for their analysis or management tasks.

Security is a key consideration in the design of the Database tab. The UTF-8 encoding check serves not only as a data integrity measure but also helps prevent potential issues with data corruption or injection. Additionally, the file upload functionality is restricted to CSV format, reducing the risk of malicious file uploads.

E. Re-train Models

A crucial aspect of the Database tab is its model retraining capability, which allows administrators to update the scam detection model as new data accumulates. This feature is essential for adapting to evolving scam tactics and maintaining high detection accuracy. When sufficient new data has been gathered, administrators can initiate the retraining process via the "**Train new dataset**" button. The system then exports the current dataset from Supabase, preprocesses the data, and trains a new logistic regression model using the most up-to-date information.

This process enables the system to learn new scam patterns, adapt to changing language, and potentially reduce both false positives and negatives. The retraining feature transforms the Database tab from a mere data repository into a vital tool for continually enhancing the scam detection system's effectiveness.

IV. Solution Implementation

The flowchart of VishGuard is shown in Fig. 6.

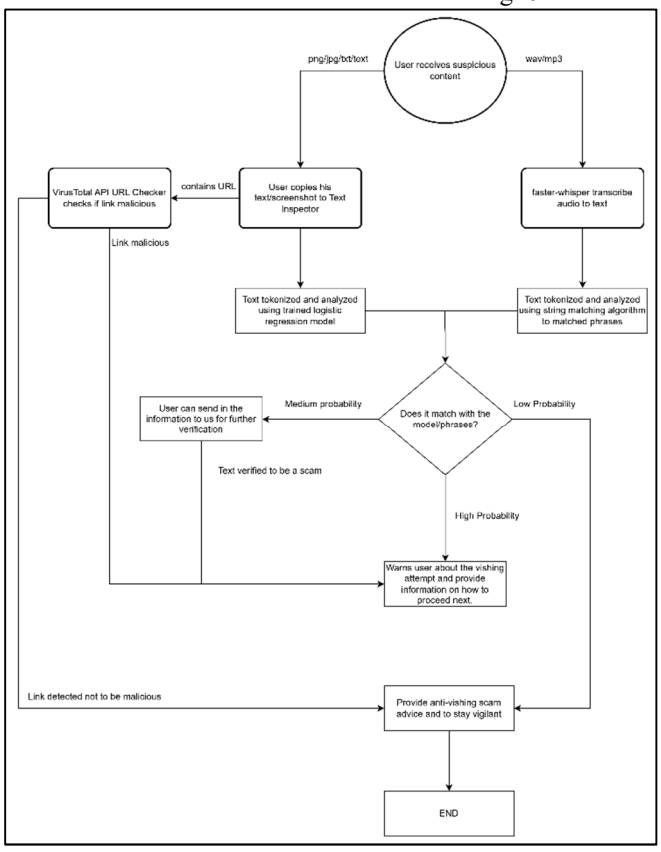

Fig. 6. Flowchart of VishGuard

A. Dataset utilised

For VishGuard, three distinct datasets were used: two for Text and Image Inspector and one for Audio Inspector.

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1) Audio Inspector

The audio inspector utilizes a dataset from [12] whereby they curated audio recordings from the FTC Project Point of No Entry initiative (PPoNE) and transcribed them [13], consisting of a total of 1432 transcribed audio texts from robocalls targeting the citizens of the United States.

2) Text and Image Inspector

Due to the different modus operandi, we had to utilize a different dataset for the scam inspector machine learning model namely, the UCI Spam Collection Dataset [14]. This dataset is particularly suitable for our scam detection model due to its comprehensive collection of spam messages, which share many characteristics with vishing and scam attempts.

The UCI Spam Collection Dataset contains many SMS messages labeled as either spam or ham, which are legitimate messages, providing a robust foundation for training a machine-learning model to identify potential scams. However, due to the age of this dataset dating back to 2012, we decided that it would be insufficient in today's digital era considering the evolution of vishing attack methods [3] hence, the need for another dataset to strengthen the machine learning model's accuracy rate.

While there are various sources of datasets available on the Internet , most of them are only available for commercial purposes and hence, to the best of our knowledge, [15] has the most comprehensive dataset consisting of only scam messages and as their dataset ranges from 2018 to 2022, it is still relevant as of today thus ensuring our model be able to combat the scams of today's world.

B. Data pre-processing

1) Audio Inspector

The data cleaning and pre-processing pipeline for the audio transcription dataset involves several key steps to refine and prepare the data for analysis. Initially, the raw dataset undergoes cleaning where irrelevant columns are removed, and only English language transcripts are retained. Duplicate transcripts and rows with null values are eliminated to ensure data quality. The cleaned dataset is then subjected to text pre-processing, which includes converting text to lowercase, removing punctuation, and tokenizing the text into individual words. Common English stop words are removed to focus on meaningful content. The script then extracts bi-grams and tri-grams (word pairs and triplets) from the preprocessed text and performs a frequency analysis to identify the most common phrases. The top 50 most frequent bi-grams and tri-grams are extracted and saved as a JSON file for future reference. This comprehensive cleaning and pre-processing approach ensure that the audio transcription data is refined, focused on English content, and prepared for effective analysis of common phrases that may be indicative of scam patterns, providing a solid foundation for subsequent string matching which will be explained in the later section.

2) Text inspector

The data preprocessing and model training pipeline for the SMS spam detection system involves a comprehensive approach to data preparation and analysis. Initially, the process combines multiple data sources, including Twitter data and the original SMS spam collection. Text files containing tweet data are processed using language detection to filter out non-English texts, ensuring dataset consistency. This preprocessed Twitter data, labeled as 'spam', is merged with the original SMS collection containing both spam and ham categories.

The combined dataset undergoes further refinement, including duplicate removal and index resetting. The cleaned data then goes through an extensive text preprocessing phase, where a custom 'clean_text' function is applied. This function tokenizes the text, removes punctuation and numbers, converts to lowercase, eliminates common English stopwords, and applies stemming using the Porter Stemmer algorithm.

The preprocessing also includes feature extraction, such as counting the number of characters, words, and sentences in each message. Exploratory data analysis is performed, generating word clouds and frequency charts for both spam and ham messages, providing visual insights into the characteristics of each category.

The cleaned text data is then vectorized using TF-IDF (Term Frequency-Inverse Document Frequency) to convert the text into a format suitable for machine learning models. Various models, namely, Logistic Regression, Decision Trees, Random Forests, Naive Bayes, K-Nearest Neighbors, and Support Vector Machines, all of which are known scam detection models, are trained and evaluated using metrics such as accuracy, precision, recall, and F1 score [16, 17]. This comprehensive preprocessing and model training pipeline ensures that the SMS spam detection system is built on a robust foundation of clean, well-prepared data and evaluated across multiple machine learning algorithms to determine the most effective approach for spam classification.

C. Data Analysis and Algorithms Used

1) Audio Inspector

The Audio detection component of VishGuard is designed to analyse audio files and detect potential scams. This functionality is crucial as scammers increasingly use diverse media formats to carry out their fraudulent activities. The primary analysis tasks performed by the audio detection component is audio transcription and analysis. The Media Inspector utilises the faster-whisper library, a high-performance variant of the Whisper model.

When a user uploads an audio file, the system first transcribes the audio content to text. The choice of faster-whisper was motivated by its ability to provide rapid, accurate transcriptions while maintaining low computational requirements. This efficiency is crucial for providing timely analysis results to users, especially when dealing with longer audio files. After transcription, the resulting text undergoes preprocessing using natural language processing techniques.

The pre-processed text is then analysed using a combination of techniques. First, the system employs a string-matching algorithm to compare the pre-processed text against a database of known scam phrases. Fig. 7 shows the top scam phrases.

The database, stored as a JSON file, contains the top 50 most frequent bi-grams and trigrams extracted from a dataset of known scam messages. The string matching is performed using the KMP algorithm, chosen for its efficiency in pattern matching, particularly when dealing with long texts or searching for multiple patterns simultaneously.

To determine if a transcribed audio message is a potential scam, the inspector uses a threshold-based approach. It counts

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the number of matches found between the pre-processed text and the known scam phrases, and calculates a total score based on the frequency of these matches in the original dataset. If either the number of matches or the total score exceeds a predefined threshold, the system flags the audio as a potential scam. This dual-criteria approach helps balance sensitivity and specificity in scam detection.


```
{
  "press 1": 848,
  "please press": 324,
  "social security": 314,
  "please press 1": 218,
  "1 speak": 209,
  "press 1 speak": 209,
  "security number": 205,
  "social security number": 199,
  "press 2": 180,
  "fraudulent activities": 167,
  "press one": 163,
  "connected concerned": 146,
  "officer press": 145,
  "concerned department": 126,
  "great day": 124,
  "amazon account": 120,
  "order connect": 120,
  "connected concerned department": 119,
  "repeat press": 116,
  "1 connected": 115,
  "1 connected concerned": 115,
  "press 1 connected": 114,
  "thank great": 112,
  "thank great day": 112,
  "authorize order": 104,
  "security administration": 102,
  "go ahead": 102,
  "social security administration": 100,
  "trying reach": 94,
  "let know": 94,
  "would like": 87,
  "get information": 84,
  "speak officer": 84,
  "officer press 1": 83,
  "minutes due": 83,
  "dont hear": 82,
  "call department": 81,
  "department inform": 81,
  "ahead get": 81,
  "go ahead get": 81,
  "order go": 80,
  "information speak": 80,
  "order go ahead": 80,
  "ahead get information": 80,
  "get information speak": 80,
  "information speak officer": 80,
  "speak officer press": 80,
  "1 thank": 79,
  "press 1 thank": 79,
  "please order": 79
}
```


Fig. 7. Top scam phrases

However, it is worth noting that the effectiveness of the audio analysis heavily depends on the accuracy of the transcription and the comprehensiveness of the scam phrase database. Unusual accents, background noise, or novel scam scripts may potentially reduce the system's accuracy.

2) Text and Image detector

As can be seen in Fig. 8., after the model was combined, our dataset's accuracy improved significantly as compared to before because having ham messages led to our original model not being able to capture many of the scam messages of today. However, the limitations of this are those messages containing URLs, even those that are legitimate, can be flagged as a false positive but this is an issue that even many models struggle till today [18].

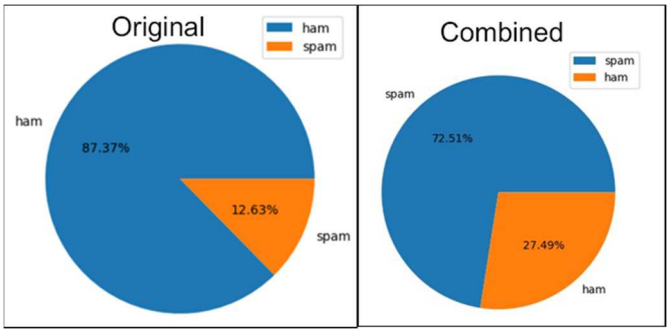

Fig. 8. Pie chart of original dataset

The system begins by processing user-uploaded images using an OCR engine, specifically Tesseract. This engine extracts any embedded text from the images, enabling further analysis. Additionally, the system utilizes the Pyzbar library for QR code detection and decoding. When a QR code is identified, it converts the QR code data into readable text, typically a URL. This extracted URL is then subjected to a safety analysis using the VirusTotal API, which aggregates results from multiple security vendors to provide a thorough assessment of potential threats, helping to identify and flag potentially malicious links.

For images that do not contain QR codes, the system focuses on analyzing the extracted text. Initially, the text undergoes preprocessing using NLTK, which includes tokenization and the removal of stop words to prepare the text for detailed examination. This preprocessed text is then scrutinized for scam indicators using the KMP algorithm. The KMP algorithm enhances text analysis by efficiently performing pattern matching. It preprocesses a list of known vishing phrases and employs a partial match table to skip redundant comparisons, making it highly effective at detecting scam-related phrases even when there are variations in the wording.

Additionally, the system features a predictive text analysis component. Users can input text for evaluation, which is analyzed for scam likelihood based on patterns and keywords. This component integrates various techniques, including a pre-trained model and heuristic rules, to assess the probability of the text being a scam. By combining these techniques, the system provides a nuanced risk assessment rather than a simple binary classification.

The integration of OCR and QR code detection with advanced text analysis algorithms like KMP, alongside machine learning models for predictive analysis, equips the system with a robust framework for detecting and evaluating potential scams. This comprehensive approach ensures effective handling of both textual and visual inputs, adapting to the evolving tactics used by scammers and enhancing overall detection capabilities.

D. Database

For in-memory sorting, the system employs pandas' sort_values() method, which uses the quicksort algorithm. Quicksort offers an average time complexity of O(n log n), making it highly efficient for large datasets. In the best case, it can achieve O(n log n), while in the worst case (rare in practice due to randomized pivot selection), it can degrade to O(n^2). This sorting approach ensures fast performance across various data distributions, particularly effective for the diverse nature of scam-related data. Data filtering operations,

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implemented through boolean indexing in pandas, exhibit a linear time complexity of O(n), where n is the number of rows; while this scales linearly with data size, it remains efficient for most practical dataset sizes due to pandas' optimized vectorized operations. The text search functionality, using pandas' str.contains() method, implements regex string matching. This operation has a time complexity that can vary depending on the complexity of the regex pattern, but approximates O(nm), where n is the number of rows and m is the average message length. While regex matching can be computationally intensive, especially for complex patterns or exceptionally large datasets, it provides powerful and flexible search capabilities crucial for identifying varied scam patterns.

V. Results and Insights

A. Accuracy Score of various Models in Scam Inspector

As shown in Fig. 9, the Support Vector Machine model has a score of 98.1%. The Random Forest Classifier achieved a score of 97.82%. The Logistics Regression model scored 97.81% for the model. While the Support Vector Machine achieved the highest accuracy, the difference between its performance and that of Logistic Regression was minimal, with only a 0.29% difference.

The Logistic Regression model, with an accuracy of 97.81%, was chosen despite its slight difference from the Support Vector Machine (98.1%) and Random Forest Classifier (97.82%) due to its advantages in simplicity, interpretability, and computational efficiency. Logistic Regression's straightforward nature allows for easy understanding and explanation of the model's results, while also being computationally less demanding and stable. These qualities make it not only a practical choice for achieving comparable accuracy but also more manageable for implementation and maintenance.

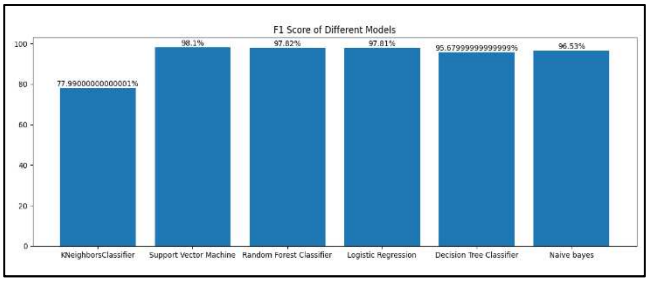

Fig.9. F1 Score of different machine models

B. Analysis of scam words in Model in Scam Inspector

Fig. 10 depicts a word cloud that highlights the most frequently used words in scam messages detected by the Scam Inspector model. In this word cloud, the size of each word indicates its frequency of occurrence in the dataset, with larger words appearing more often in scam messages. The visualization highlights several key characteristics of scam messages beyond mere frequency. Key observations from the word cloud include:

1) *Spelling Errors:* The presence of words such as "pleas" instead of "please" suggests deliberate misspellings. Scammers often use such errors to bypass spam filters that are designed to catch common scam phrases [19].

2) *Choice of words:* Words like "dear" and "please" are chosen for their ability to manipulate the recipient. "Dear" is typically used in the greeting of emails, such as "Dear Mdm" or "Dear Sir," to create a sense of formality and trust. "Please" is employed to create a sense of urgency, urging the recipient to take immediate action.

3) *Formal Yet Suspicious Language:* The combination of formal address ("dear") with urgent language ("please") is a common tactic to create a sense of legitimacy while pressing the recipient to act swiftly without overthinking.

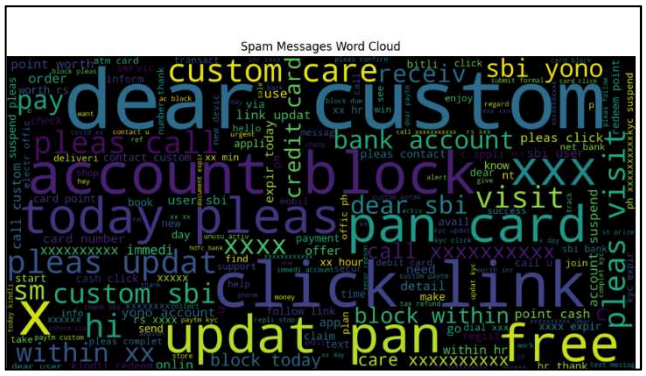

Fig. 10. Word Cloud of the most used words

This analysis of the word cloud provides insights into the strategic choices made by scammers in their language use, emphasizing the importance of detecting not just the words themselves but the patterns and intent behind them.

C. Top Scam Phrases in Media Inspector

Fig. 11 displays a bar chart representing the top 30 most frequently used words in scam messages identified by the Scam Inspector model. The most popular scam phrases are 'http,' 'pleas' and 'dear.' The phrase 'http' has appeared close to 5000 times while 'pleas' which appeared slightly above 3000 times.

This highlights the deliberate choices made by scammers to bypass filters and trick recipients. This understanding is crucial for improving the detection capabilities of the Scam Inspector model, making it more robust and effective in identifying and mitigating scam messages.

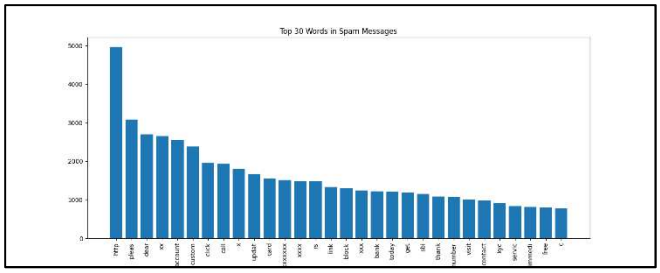

Fig. 11. Bar chart of top 30 words in spam messages

VI. Future Work

A. Dataset

Although dataset in [14] ranges from 2018 to 2022 which is still relevant as of today, one of our future works will explore and use more recent and diverse datasets to improve the system's accuracy, particularly in combating new forms of scams.

B. Performance Evaluation

Currently we use the Logistic Regression model in the VishGuard with the accuracy of 97.81%. Our future work will use real-world data to demonstrate VishGuard's

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effectiveness, show the system's real-time performance under heavy load, and compare performances among VishGuard, ScamShield and CheckMate.

C. User Experience

Our future work may invite students and elderly people to use VishGuard. Based on their feedback, we will further improve VishGuard for easy use, especially for non-technical users.

D. Adoption of VishGuard

After completing the above works, we may present our VishGuard to the policymaker for the potential adoption to effectively defend against vishing attacks.

VII. Conclusion

In conclusion, VishGuard represents a significant advancement in the battle against digital scams, particularly vishing attacks. By integrating multi-modal analysis capabilities, advanced NLP techniques, and machine learning algorithms, this web-based platform offers a comprehensive and adaptive solution to the evolving landscape of cyber threats.

Its ability to process text, audio, and images, coupled with a user-friendly interface and continuous improvement mechanism, sets VishGuard apart as a versatile and powerful tool. While the system shows great promise, challenges remain, such as the need for more localized data and the potential for false positives in URL detection. Future developments could focus on expanding the dataset, implementing more advanced machine learning models, and integrating with real-time communication platforms. As digital threats continue to evolve, VishGuard and similar innovative platforms will play a crucial role in safeguarding individuals and organizations in our increasingly complex digital world.

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